

Cloud Native Streaming Platform Netflix-inspired

Santiago Antonio Gualteros Bustos

Nicolas Castro Rivera

Andrés Julian Vargas Medina

Erick Santiago Buitrago Peña

1 Executive Summary

Netflix is a streaming platform that provides this service for subscriptions. Whose vision is entertaining customers through series and movies. This content can be accessed around the world by millions of customers those who want to watch their favorites shows at any time. For that reason is an application demands great availability, good performance with heavy content such as multimedia and personalized experience with assertive recommendations.

2 System Description

Business problem, actor catalog, use case definitions, and system boundary diagram

2.1 Business Problem

Netflix solves the problem of providing a large amount of audiovisual content to millions of users through the internet while giving each user a convenient and personalized experience.

Users want to watch movies and series whenever and wherever they want without having to download the entire content beforehand. At the same time, Netflix needs to help users find something they want to watch among a very large catalog.

The platform provide another services that not will be consider in this project like: Option for subtitles, interactive experiences or content preview.

2.2 Actors

The application has 4 users represented by a person, each of this person has a particular function and use the application for different purposes. Also there are some system actor that provide services for the application.

Actor	Role	Frequency of interaction	Expectation
Customer	A person who has a monthly membership plan with the platform.	High	Be able to watch series and movies without interruptions and receive relevant recommendations.
Operational user	A member of the operations team responsible for managing content within the system.	Medium	Be able to manage all the content without problems.
Data analyst	A member of the Data & Insights area who can access data and statistics.	Medium	Have access to useful data with clear statistics.
Platform administrator	Is responsible for managing users and the business model.	Low	Be able to manage the users of the system and membership configurations.
Media Convert service	It allows videos to be sent in small segments.	High	The user does not need to download the entire video; it is stored in fragments in the device buffer.
CDN service	Servers that help reduce latency by delivering content from the nearest server.	High	Ensure rapid response to high volumes of requests.
Storage	Storage for multimedia content and metadata, useful for performing queries.	High	Persist all data over time.

Table 1: Actors, roles, interaction frequency, and expectations of the Netflix platform.

2.3 Main Use Cases

Subsection example 2

2.4 System Boundaries

Subsection example 2

3 Functional Requirements

This requirements specify what application does. There are 11 functional requirements divided in domain

Table 2: System Requirements Matrix Categorized by Domain / Bounded Context

ID	Requirement (The system shall...)	Inputs	Outputs	Domain / Context
RF - 01	Allow customer registration through paid membership.	Email, password, payment details	Customer account creation notification	Subscription Management (Subscription)
RF - 02	Allow membership payment processing using online payment methods.	Payment method info (card/gateway)	Payment confirmation or error message	
RF - 05	Allow customers to cancel their active membership subscription.	Cancellation request and reason	Membership cancellation status confirmation	
RF - 03	Allow any user to authenticate using email and password credentials.	Email and password credentials	Access granted to appropriate user dashboard	Identity & Access Management
RF - 04	Allow users to update their access credentials.	Registered email, current and new credentials	Credential update success or failure notice	(IAM)
RF - 06	Allow any user to update their personal profile information.	Authenticated session, updated attributes	Updated profile status or error message	
RF - 10	Allow platform administrators to manage user accounts and roles.	Admin request parameters and action payload	Success/failure message and updated user state	
RF - 07	Allow customers to stream movie or episode content instantly without prior downloading.	Chapter/movie selection	Instant video playback stream	Streaming Core
RF - 08	Provide personalized and timely content recommendations to customers.	Contextual data, watch history, search queries	Personalized content feed on main screen	Recommendation Engine
RF - 09	Allow operations personnel to manage media content catalog items.	Metadata creation, update, query, or deletion requests	Action result message and updated catalog state	Content Management
RF - 11	Allow data analysts to query and view streaming usage analytics.	Time range and metric filter parameters	Processed analytical reports and metrics	Data & Insights
RF - 12	Allow customers to search for movies or series in the catalog.	Search query string	List of matching movies/series and metadata	Catalog & Discovery

4 Non-Functional Requirements

Numbered catalog of measurable non-functional requirements per category

5 Workload Model

Structured table with all workload parameters and justification for each value

References

- [1] Rummery, G. A., & Niranjan, M. (1994). On-line Q-learning using connectionist systems. *Technical Report*, Cambridge University Engineering Department.